Atharva Mohite

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EDUCATION

Georgia Institute of Technology — *Aug 2026 – May 2028 (Expected)*
Master of Science in Computer Science, Machine Learning — Atlanta, GA
Relevant Coursework: Machine Learning, Big Data Systems, Data Visualization and Analytics

Sardar Patel Institute of Technology — *Aug 2019 – May 2023*
Bachelor of Technology (BTech) in Computer Science | **CGPA**: 9.39/10 — Mumbai, India

EXPERIENCE

JPMorgan Chase — *Jan 2026 – Aug 2026*
Software Engineer II

- Owned cross-service design and production readiness for a new business line across **7 microservices** on a multi-tenant settlement platform, scaling throughput to **5M+** trades per day.
- Built an **agentic SDLC** pipeline that autonomously translated Jira requirements into code changes across a multi-service architecture, reproduced issues with local infrastructure, validated via automated tests, and raised developer-ready PRs for review, with **75%** PRs merged with minimal edits.
- Built a regression framework using **LLM agents** that generates and diagnoses multi-service test workflows, automating **300+** scenarios and cutting manual test effort by **90%**; adopted by 30+ engineers across 5 teams.
- Designed an aggregation engine in Spring Boot reducing high-volume trade streams into consolidated settlement instructions across configurable dimensions, cutting down instruction volume by **70%**.

JPMorgan Chase — *Feb 2023 – Dec 2025*
Software Engineer I

- Led extraction of a high-throughput microservice from a legacy trade settlement monolith, processing **1M+** messages per day, with an event-driven, non-blocking stack with Kafka and Spring WebFlux.
- Designed distributed end-of-day orchestration coordinating 3 services to close and settle the day's book within a **30**-minute cutoff, handling race conditions, ordering dependencies, and concurrent state transitions with idempotent replay on failure.
- Architected a materialized datastore using **Elasticsearch**, aggregating and rehydrating data from multiple source systems to provide **sub-150ms** complex search over **750k** records, enabling the decommission of **4** legacy systems.
- Migrated 15 legacy applications to an in-house **Kubernetes** platform, saving **$450k** annually in infrastructure costs and reducing deployment time by **50%.**

JPMorgan Chase — *Jan 2022 – Jul 2022*
Software Engineer Intern

- Developed a cloud resource discovery framework that helped discover **10k+** resources, consolidating AWS Config, Splunk, Terraform, and tagging APIs into a unified inventory using Qlik Sense dashboards for tracking resource ownership, enabling data-driven cleanup decisions and improving infrastructure visibility for the DevOps team.

SKILLS

Languages: Python, Java, TypeScript, SQL
AI/ML: LLM agents, RAG, LangChain, LangGraph, MCP, vector DBs, scikit-learn, pandas, evals and prompt design
Backend & Infra: Spring Boot, Flask, Node.js, Django, Relational Databases (Oracle), Elasticsearch, AWS, Terraform, Kubernetes, Docker, Jenkins, Shell Scripting, React, Next.js
Other: System design (scalable and distributed systems), Event-driven architecture (Kafka), RESTful and GraphQL API design, Cloud Architecture, Testing (unit, integration, E2E), CI/CD pipelines, Design Patterns, Monitoring and Observability (Splunk, Grafana).

PROJECTS

Canary | Docs | Python, scikit-learn

- Cut coding-agent compute by **25%** with a one-point drop in solve rate by predicting run failure from the first few steps of execution.
- Built supervised learning models over 80k coding-agent trajectories from Agent SWE Bench to predict failure from the first few steps, achieving **0.80** PR-AUC on a task-controlled evaluation.
- Combined supervised prediction with unsupervised clustering to identify recurring failure modes across 59k failed runs, explain why agents fail, and determine when an agent should continue or stop.

RESEARCH

Leveraging LLMs for Video Querying — 2023
IEEE 7th International Conference on Computer Applications in Electrical Engineering-Recent Advances (CERA) — Paper Link

- Built a long-video search pipeline that converted surveillance footage into time-stamped captions with BMT and Vid2Seq, then used GPT-3.5, GPT-4, and LLaMA to retrieve events from natural-language queries.
- Benchmarked 12 action-recognition and keyframe-extraction combinations; the best result reached **84.53%** average accuracy, while keyframe selection reduced reported per-epoch time by up to **68.75%**.

SecurePark - a vehicle intrusion detection system — 2022
IEEE Asian Conference on Innovation in Technology (ASIANCON) — Paper Link

- Trained a YOLOv5s license-plate detector on labeled training images and integrated OCR, vehicle authorization, alerts, saved recordings, and usage analytics into a working monitoring dashboard.